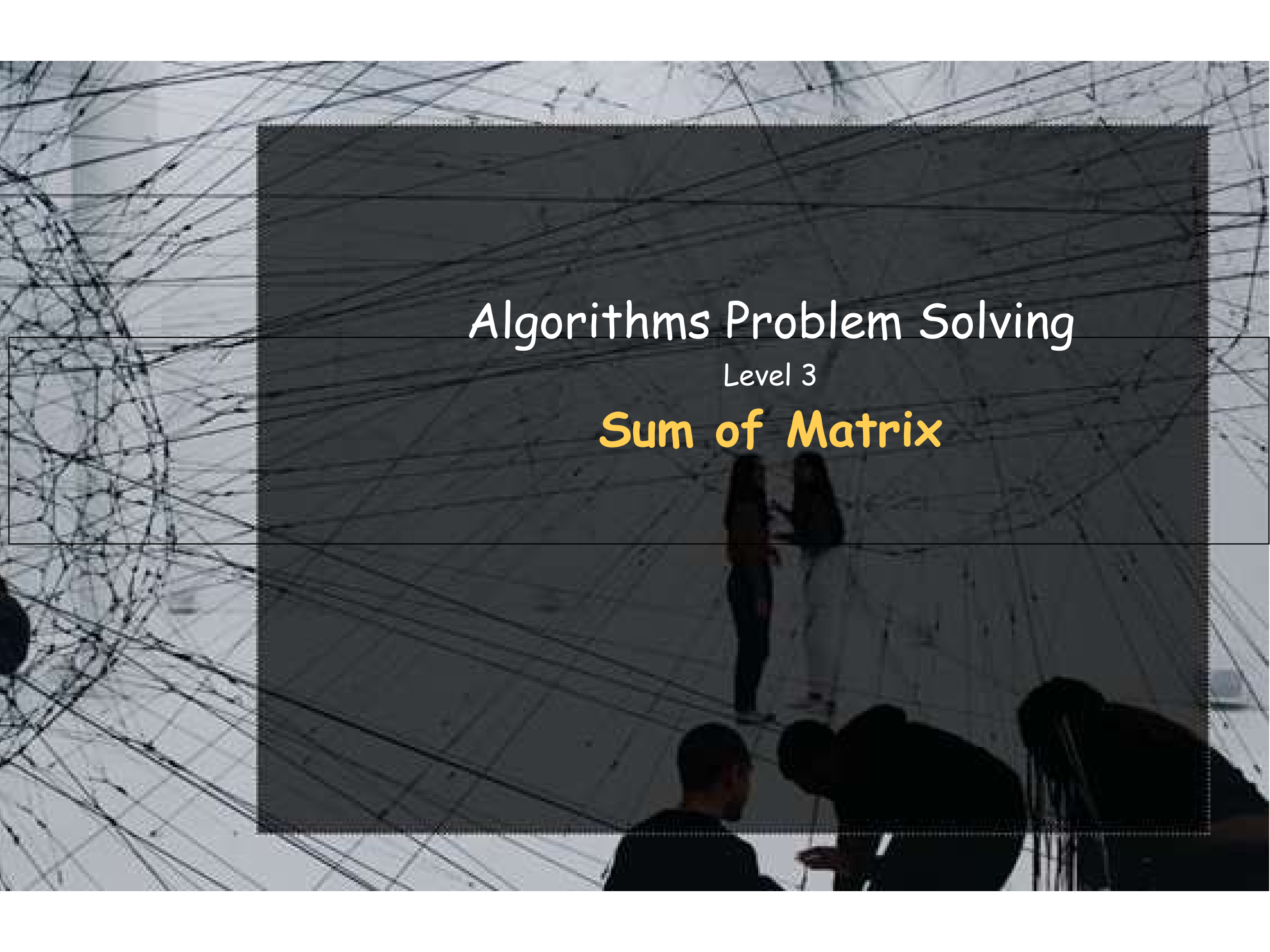


Algorithms Problem Solving

Level 3

Sum of Matrix



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Definition

File-Based Record Deletion

 Program deletes a client identified by Account Number from a text-based database.

 Because variable-length text records cannot easily be deleted in place, the remaining records are written back to the file.

Explanation

Target Search and Confirmation

-  The program first locates the client using its Account Number.
-  The user is asked to confirm the deletion before modifying persistent data.

Deletion Marking

-  A `MarkForDelete` flag can be set on the matching in-memory record.
-  The flag allows the saving process to skip that record during rewriting.

File Rewriting

-  The file is opened in a mode that allows its previous content to be replaced.
-  All records except the marked record are serialized and written back.

Persistent Result

-  After rewriting, the deleted client no longer appears in the file.
-  The operation therefore changes the persistent dataset, not just the in-memory vector.

Procedures

1. Load all clients from the file.
2. Search for the target Account Number.
3. Display the found client.
4. Ask the user to confirm deletion.
5. Mark or remove the selected record.
6. Rewrite the file using only remaining records.
7. Display a success message.

💡 Example

Enter Account Number to Delete:

A150

Are you sure you want to delete this client? Y

➡ Result

Client Deleted Successfully!

Note

- ◆ Rewriting is a common technique for deleting records from simple text-based databases.
- ◆ The original file contents are replaced by the filtered record set.
- ◆ Confirmation helps prevent accidental permanent deletion.

⊘ Warning

- ◆ Once the file is rewritten, the deleted record is no longer present unless a backup exists.
- ◆ Opening or saving the wrong filename can lead to unexpected data loss or an apparently empty database.
- ◆ The deletion process should operate on freshly loaded data to avoid overwriting newer changes with stale in-memory records.



Comparison

▨ Mark-and-Rewrite Deletion

- ✓ Marks the target record in memory.
- ✓ Rewrites the file while skipping marked records.
- ✓ Works naturally with variable-length text records.

▤ Direct In-Place File Editing

- ✓ Attempts to modify the existing file without rebuilding it.
- ✓ Is difficult for variable-length text records.
- ✓ Usually requires more complicated file manipulation.

✦✦ Summary

- ✓ Finds the client using its Account Number.
- ✓ Requests explicit confirmation.
- ✓ Marks or removes the selected record in memory.
- ✓ Rewrites the file without the deleted record.

Thank you 😊